

# Academic Review Report — Round 1

## “Can Anything Beat VIX? A Systematic Out-of-Sample Evaluation of Eleven Signal Families for Equity Volatility Forecasting and Volatility Timing”

Document type: Journal paper (targeting *Journal of Forecasting*)

Version audited: `main_v2.tex`

Review date: 2026-04-05

Reviewer: Claude Opus 4.6 (1M context)

Standard: Strict (top-tier submission readiness)

## Review Summary

|                             |                                              |
|-----------------------------|----------------------------------------------|
| Severe issues (must fix)    | 4 items                                      |
| Medium issues (should fix)  | 8 items                                      |
| Minor issues (optional fix) | 6 items                                      |
| Overall assessment          | <b>Good — publishable after R1 revisions</b> |

**Overall impression.** This is a well-structured, clearly written paper with a genuinely useful contribution: a systematic horse race of 11 signal families under a unified evaluation pipeline with strict methodological safeguards. The “informative null” framing, the insurance interpretation of VT, and the criterion-dependent model ranking analysis are all strong contributions. The paper is close to submission-ready but has four severe issues—one data-integrity problem and three conceptual/framing problems—that must be addressed before submission.

## 1. Logic Structure

The overall argument flows well: motivation → VIX as benchmark → data/signals → methodology → results → interpretation → conclusion. The dual- $R^2$  framework (Campbell-Thompson vs. VIX-relative) is a strong pedagogical choice that clarifies why signals can “beat the mean” yet add nothing beyond VIX.

**M1: Tension between “time-invariant” claim and era exceptions.** The paper claims VIX sufficiency is “time-invariant” (Abstract, Table 5 notes, Table 6 notes), yet the data shows this is not perfectly true. The coefficient of variation of 0.33 is modest but not zero; more importantly, the competing-signal analysis in Table 6 is where the contradiction becomes severe (see S1 below). The paper should distinguish between the VIX–RV *relationship* being time-stable (true, with CV=0.33) and competing signals having *zero incremental power in every era* (false, per the K752 data).

**Strengths.** The six methodological safeguards (pre-specification, lag enforcement, TX costs, multiple testing, proxy-robustness, cross-era) are clearly enumerated and consistently applied. The Proposition 1 insurance framework is a nice formal contribution.

## 2. Research Motivation and Contribution

**Strong.** The research question is important and well-motivated. The connection to the existing horse-race tradition (Hansen & Lunde, 2005) and the gap identification (no prior study tests 11 signal families simultaneously) are convincing. Three distinct contributions (informative null, insurance framework, criterion-dependent rankings) are well-delineated.

**M2: “Pre-registered” language.** The paper uses “pre-registered horse race” in the introduction, but there is no evidence of formal pre-registration (e.g., on OSF or AEA RCT Registry). The signals were “pre-specified” before examining out-of-sample results, which is good practice but not the same as pre-registration. Suggest changing “pre-registered” to “pre-specified” consistently to avoid reviewer pushback. (Checked: line 74 says “pre-registered.”)

### 3. Literature Review

**Adequate.** The paper covers the key literature: Poon & Granger (2003) survey, Christensen & Prabhala (1998), Jiang & Tian (2005), Bekaert & Hoerova (2014), Hansen & Lunde (2005), Patton (2011), Moreira & Muir (2017), and relevant signal-specific studies.

**M3: Missing recent literature.**

- No discussion of **Bekaert, Engstrom, & Ermolov (2021)** who decompose VIX into risk and uncertainty components — directly relevant to the VRP discussion.
- No mention of **Feunou, Jahan-Parvar, & Okou (2018)** on downside/upside variance risk premia — relevant to Family 4.
- **Bali, Brown, & Caglayan (2014)** on economic uncertainty and portfolio allocation — relevant to the insurance framework.
- The HAR-RV literature post-Corsi is thin: should cite **Bollerslev, Patton, & Quaedvlieg (2016)** on realized semi-variance modeling.

### 4. Model Specification

**Sound.** The VIX benchmark regression (Eq. 4), incremental test (Eq. 5), and strategy construction (Eq. 7) are standard and appropriate. The QLIKE loss function and MCS procedure are correctly specified.

**M4: No formal definition of “time-invariance.”** The paper adopts “ $CV < 0.50$ ” as a threshold for time-invariance (Table 5 notes) but does not justify this cutoff. Is 0.50 standard in the literature? A formal structural break test (e.g., Bai-Perron) or a sup- $F$  test on the VIX–RV slope across eras would strengthen the claim substantially.

### 5. Equations and Derivations

**Correct.** The VIX formula (Eq. 1), VRP decomposition (Eq. 2), RV definition (Eq. 3), benchmark regression (Eq. 4), DM test (Eq. 6–7), strategy rule (Eq. 8), TX cost (Eq. 9), information aggregation (Eq. 10–11), and insurance proposition (Eq. 12) are all mathematically correct.

Minor: The Proposition 1 derivation (Eq. 12) assumes quadratic utility approximation to CRRA, which is valid for small risk but should be noted as an approximation.

### 6. Symbol Consistency

No symbol conflicts detected.  $\sigma^2$ ,  $h_t$ ,  $r_t$ ,  $RV$ , and VIX are used consistently throughout.

## 7. Research Methodology

**Strong overall.** The unified pipeline, lag enforcement, TX accounting, and multiple-testing framework are exemplary.

**M5: Families 5–7 receive weaker treatment.** Families 5 (multi-asset optimization), 6 (ERC), and 7 (Bitcoin) lack formal DM tests and OOS  $R^2$  metrics because they are “portfolio strategies without direct volatility forecasting components.” This is understandable but creates an asymmetry in the evaluation. Consider either (a) adding return-based DM tests for these families, or (b) explicitly noting in Table 2 that these families test a different hypothesis (portfolio construction vs. volatility forecasting).

**M6: Google Trends weekly-to-daily conversion.** Family 9 forward-fills weekly data to daily frequency, which mechanically inflates serial correlation and may bias the DM test (which uses HAC standard errors with bandwidth 22). This should be discussed as a limitation. Consider also showing results at weekly frequency to verify that the null is not an artifact of the conversion.

## 8. Data and Sample Design

**Good.** The 8,325-day sample (1993–2026) is substantial. Era definitions are reasonable and non-overlapping. Data sources (Yahoo Finance) are transparent.

**M7: Heterogeneous sample periods.** Individual signals range from 797 observations (Google Trends) to 8,325 (Calendar). Cross-signal comparisons in Table 2 are conducted over different samples, which limits direct comparability. The paper should acknowledge this more explicitly and consider whether results change when restricted to the common sample (2011–2026, approximately).

## 9. Expected Results and Claims

The “informative null” framing is well-executed. The insurance framework provides practical value. The criterion-dependent ranking analysis is a genuine secondary contribution.

**M8: HAR-RV 41.8% improvement claim needs qualification.** The 41.8% QLIKE improvement for HAR-RV is based on K745, a pilot study with only  $N = 37$  observations. The paper qualifies this as “preliminary,” but it is mentioned three times (Abstract, Section 7.1, Section 8) and may leave an outsized impression. Either (a) expand the pilot to a full study before submission, or (b) reduce prominence to a single mention in the conclusion with an explicit “ $N = 37$ ” caveat.

## 10. Citation Verification (Cross-Referenced with Citation Report)

See the companion `citation_check_v1.md` for full details. Key findings:

- 40 total citations; 36 verified, 3 with minor issues, 1 with an error.
- **Luo & Zhang (2019):** The paper cites this as supporting the VIX term structure regime indicator claim. However, the 2019 paper on “VIX term structure and VIX futures pricing with realized volatility” in *Journal of Futures Markets*, 39(1), 72–93 is by **Huang, Tong, & Wang**—not Luo & Zhang. Luo & Zhang published “The Term Structure of VIX” in the same journal in **2012** (vol. 32, no. 12, pp. 1092–1113). The bibliography entry needs correction.
- Several citations lack DOIs (see citation report).

## 11. Number Verification (Cross-Referenced with Audit Report)

The companion audit (audit\_step1\_2.md) and reproduce report (reproduce\_report.json) verify 93 of 100 checked values. Tables 1, 3, 4, 5, 7, 8, 9 all match their source experiments. The critical finding concerns Table 6.

### Detailed Issue List

#### Severe Issues (MUST FIX)

##### S1. Table 6: Competing Signals by Era — Fabricated or Erroneous Values.

*Location:* Table 6 and its notes (lines 573–593).

*Problem:* The paper reports uniformly tiny incremental  $R^2$  values (all  $< 0.001$ ) for three competing signals across five eras, and the table note states “No signal passes the Harvey (2016)  $|t| > 3.0$  threshold in any era.” However, the source experiment K752 (part\_d\_competing\_signals\_by\_era) shows:

- **Era 3 (GFC):** Overnight VIX incremental  $R^2 = 0.0039$  (paper: 0.0004); VRP = 0.016 (paper: 0.0008); Vol Momentum = 0.0216 (paper: 0.0006). Harvey-passing  $|t|$ -statistics: Overnight  $|t| = 3.15$ , VRP  $|t| = 6.51$ , Vol Mom  $|t| = 7.60$ . **All three pass Harvey in GFC.**
- **Era 5 (COVID):** Vol Momentum incremental  $R^2 = 0.0372$  (paper: 0.0002); Overnight VIX = 0.0032 (paper: 0.0003). Vol Mom  $|t| = 9.30$  (**passes Harvey**).

The paper’s values for Eras 3 and 5 are **orders of magnitude too small** and appear to be from a different calculation or a data-entry error. The table note’s claim “No signal passes Harvey in any era” is **false** per the K752 data.

*Impact:* This is the paper’s most serious problem. VIX sufficiency has **era-specific exceptions** during high-stress periods (GFC, COVID). This does not destroy the paper’s main finding (the full-sample null is robust), but it requires honest acknowledgment.

*Required fix:* (a) Correct Table 6 values to match K752. (b) Change the table note to: “Three signals pass the Harvey threshold during the GFC era; one passes during the COVID era. No signal passes in the remaining three eras.” (c) Add a paragraph discussing why crisis-era exceptions exist (extreme VIX levels create mechanical relationships with VRP and overnight changes) and why they do not undermine the full-sample conclusion.

##### S2. The “36 VIX Sufficiency Tests” Count is Unverifiable.

*Location:* Abstract (line 85), Conclusion (line 820).

*Problem:* The paper claims “36 VIX sufficiency tests” and “36 null results accumulated across this research program.” The audit found that the knowledge-base numbering is inconsistent: numbers #34–#40 appear in early experiments (K535–K638), then the sequence restarts at #25 with later experiments (K863–K879). Duplicate numbers exist (e.g., two #36). The exact count of 36 cannot be verified.

*Required fix:* Either (a) provide a definitive appendix listing all 36 tests with experiment IDs, or (b) change to a defensible statement such as “more than 30 VIX sufficiency tests.”

##### S3. 12/VIX Numerator Description is Incorrect.

*Location:* Section 5.1, line 377: “The numerator 12 approximates long-run average VIX.”

*Problem:* The long-run average VIX in the sample is 19.5 (K752 full-sample mean). The number 12 does **not** approximate long-run average VIX. Rather, 12 represents a target annualized volatility level of 12%, which determines the equity allocation via the ratio 12/VIX. When VIX equals its mean ( $\sim 19.5$ ), the equity weight is  $12/19.5 \approx 61\%$ , not 100%.

*Impact:* A reviewer familiar with VIX would immediately spot this error, undermining credibility.

*Required fix:* Change to: “The numerator 12 represents a target annualized equity volatility of 12%, calibrated to produce conservative equity exposure under typical VIX levels.” or similar.

#### **S4. Citation Error: Luo & Zhang (2019).**

*Location:* Section 3.2 (line 206), Bibliography (lines 921–922).

*Problem:* The paper cites “Luo & Zhang (2019)” for the VIX term structure regime indicator and lists the bibliography entry as: “Luo, X., & Zhang, J. E. (2019). VIX term structure and VIX futures pricing with realized volatility. *Journal of Futures Markets*, 39(1), 72–93.” However, the actual 2019 paper with that title in *JFM* 39(1):72–93 is by **Huang, Tong, & Wang** (DOI: 10.1002/fut.21955). Luo & Zhang published “The Term Structure of VIX” in *JFM* 2012, vol. 32, pp. 1092–1113 (DOI: 10.1002/fut.21572), and “Instantaneous Squared VIX and VIX Derivatives” in *JFM* 2019, vol. 39, pp. 1193–1213 (DOI: 10.1002/fut.22037). The cited title, journal, volume, and pages belong to a different paper by different authors.

*Required fix:* Either (a) correct the citation to Huang, Tong, & Wang (2019) if that is the intended source, or (b) change to Luo & Zhang (2012) and update the bibliography entry accordingly.

### **Medium Issues (SHOULD FIX)**

**M1.** Tension between “time-invariant” claim and era exceptions (see Section 1 above).

**M2.** “Pre-registered” language without actual pre-registration (see Section 2 above).

**M3.** Missing recent literature — Bekaert, Engstrom, & Ermolov (2021); Bollerslev, Patton, & Quaadvlieg (2016); Feunou et al. (2018) (see Section 3 above).

**M4.** No formal definition or test of “time-invariance” — the  $CV < 0.50$  threshold is ad hoc (see Section 4 above).

**M5.** Families 5–7 lack formal DM tests, creating asymmetry (see Section 7 above).

**M6.** Google Trends weekly-to-daily forward-fill may bias serial correlation (see Section 7 above).

**M7.** Heterogeneous sample periods across signals limit comparability (see Section 8 above).

**M8.** HAR-RV 41.8% claim based on  $N = 37$  pilot; mentioned 3 times (see Section 9 above).

### **Minor Issues**

**m1.** Table 2, Family 3 (Behavioral Sentiment): The IS  $t$ -stat of 1.64 appears to come from the DM OOS statistic ( $K732: dm\_stat\_oos = 1.637$ ), not from the in-sample BSI coefficient  $t$ -stat ( $K732: bsi\_t\_stat = 5.58$ ). The DM  $|t|$  column shows 0.52, but  $K732$ ’s  $dm\_bsi\_vs\_bh = -1.18$ . These may come from a different regression not stored in the JSON. Clarify the source or add the intermediate computation to the experiment results.

**m2.** Table 2, Family 7 (Bitcoin): The “Partial  $r|VIX$ ” of 0.178 appears to be a simple bivariate correlation ( $full\_sample\_VIX\_BTC\_RV\_corr$ ), not a partial correlation controlling for VIX. If it is a partial correlation of BTC RV with SPY RV controlling for VIX, this should be clarified.

**m3.** Table 2, Families 1, 8, 10: Several “Partial  $r|VIX$ ” values are not directly traceable to stored JSON fields. They should be saved to the experiment results for full replicability.

**m4.** Insurance drag discrepancy:  $K828$  (dedicated SPY-only experiment) shows 4.17%/yr for 12/VIX drag, while  $K738$  (cross-asset) shows 3.49%/yr. The paper uses 3.49%. Consider noting whether this is SPY-only or cross-asset average, since the distinction matters for investors.

**m5.** The abstract is 319 words—on the long side for most journals. Consider trimming to  $<250$  words if the target journal requires it.

**m6.** Several bibliography entries lack DOIs. See citation report for the full list.

## Summary and Revision Priority

**Overall assessment:** This is a strong paper with a clear contribution. The research design (unified pipeline, 11 families, strict controls) is excellent. The insurance framework and criterion-dependent rankings are genuine secondary contributions. After addressing the four severe issues, this paper should be competitive at *Journal of Forecasting*, *International Journal of Forecasting*, or similar venues.

### Revision priority (highest to lowest):

1. **S1:** Correct Table 6 values and acknowledge era exceptions — this is a data-integrity issue that could sink a paper at review.
2. **S3:** Fix the 12/VIX numerator description — factual error visible to any volatility researcher.
3. **S4:** Correct the Luo & Zhang (2019) citation — wrong authors for the cited paper.
4. **S2:** Either substantiate or soften the “36 tests” claim.
5. **M1:** Reconcile “time-invariant” language with era exceptions.
6. **M4:** Add a formal time-invariance test or justify the CV threshold.
7. **M8:** Reduce prominence of HAR-RV preliminary result.
8. **M2:** Change “pre-registered” to “pre-specified.”
9. **M3, M5, M6, M7:** Literature gaps, asymmetric treatment, data-frequency issues.
10. Minor items (m1–m6).

**Estimated revision effort:** 2–3 days for S1–S4 and M1–M8; minor items can be done in parallel.